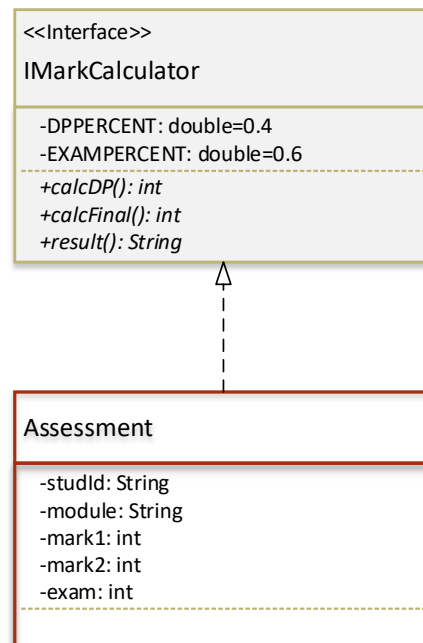


## Question 1

Class **Assessment** implements the interface **IMarkCalculator**. **Assessment** contains `studId` (student number), `module` (name of module), `mark1` and `mark2` (test marks) and `exam` (exam mark).

**IMarkCalculator** contains constants and methods for calculating the dp and final mark, and to determine whether a student passed or failed (based on the final mark).

`DPPERCENT` and `EXAMPERCENT` are constants (40% and 60% respectively). The method `calcDp()` returns the dp (average of test marks); `calcFinal()` returns the final mark as  $(DPPERCENT \times dp) + (EXAMPERCENT \times exam)$  and `result()` returns "Passed" if the final mark is greater than 50; otherwise "Failed".



1. Write the code for the **IMarkCalculator** and **Assessment**.
2. Write **Tester** to display the test marks, exam, dp, final exam mark and result for a student. A sample output is shown below.

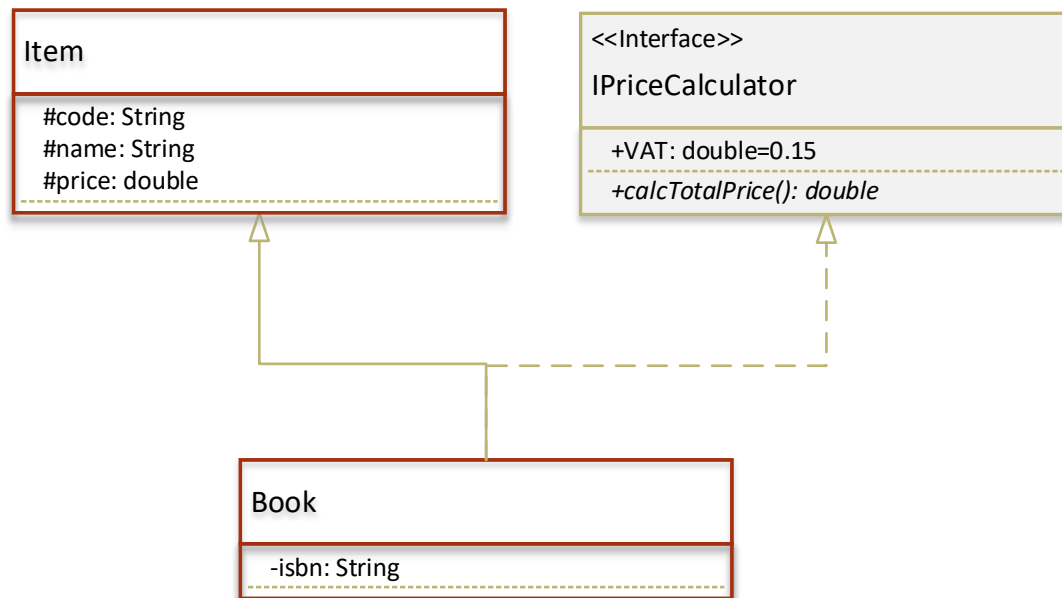
```
StudId=20202020      Module=IPRT
Mark 1: 50      Mark 2: 50      Exam: 50
Dp: 50      Final: 50
Result: Passed
```

3. Can you create an instance of **IMarkCalculator**? Explain why.
4. Modify the **Tester** as follows: add a statement which assigns an instance of **Assessment** to **IMarkCalculator** variable (**im**) as shown below. Is this statement valid? Why?

```
IMarkCalculator im=new Assessment();
```

## Question 2

The diagram shows the classes **Item** and **Book** and the interface **IPriceCalculator**. VAT is a constant (15%); the method *calcTotalPrice()* returns the total price of an item (i.e., price + Vat payable on the item).



1. Write the code for **IPriceCalculator**, **Item** and **Book**
2. Write **Tester** to display the code, name, isbn, price and total price of a book. A sample output is shown below.

```
Code:Prog      Name:Java for Dummies      Price=R657.0
Total Price=R755.55
Isbn:978-1-119-23555
```

### Question 3

Write the code for IPriceCalculator, Item, Book and Phone. Write a class Tester to test the programs

1. Write a tester to test the pr

